

Student Performance Analysis Report

Academic performance overview, learner segment insights, and data-driven recommendations

100 Students	61.18 Overall Average	41% Test Prep Completion
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Tools Used: Excel / Power BI / DAX / Data Visualization / Business Reporting

Report Objective

This report summarises a two-page Student Performance Dashboard and provides a business-friendly interpretation of the dataset, dashboard visuals, performance patterns, and recommended actions for improving learning outcomes.

Dataset Scope

The analysis covers 100 student records across gender, ethnicity, parental education, test preparation status, and three subject scores: Math, Reading, and Writing.

Executive Summary

The student dataset shows moderate academic performance, with an overall average score of 61.18 out of 100. Subject averages are tightly grouped: Math is highest at 62.51, Writing is 60.53, and Reading is 60.49. This indicates that no single subject dominates performance; the academic pattern is broadly balanced but leaves room for improvement across all three subject areas.

The strongest performance pattern in the analysis is the effect of test preparation. Students who completed test preparation achieved an average score of 66.34, compared with 57.59 for students who did not complete test preparation. This represents an 8.75-point performance advantage, making test preparation the clearest intervention opportunity in the dataset.

The dashboard also shows that most students fall in the 60-69 score band, while 10 students fall below 50. This suggests that improvement efforts should not only target the lowest-performing students, but also help the large mid-performing group move into higher achievement bands.

62.51 Math Avg	60.49 Reading Avg	60.53 Writing Avg	+8.75 pts Prep Advantage
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Key Findings

- The overall average score is 61.18, which suggests a moderate performance level with clear room for academic improvement.
- Only 41% of students completed test preparation, yet this group scored 8.75 points higher on average than students who did not complete preparation.
- Math is the strongest subject for the largest number of students, appearing as the top subject for 42 students.
- Reading appears most often as the weakest subject, appearing as the lowest-scoring subject for 41 students.
- Parental education groups show a performance gap of about 4.13 points between the highest and lowest group averages.
- The Reading vs Writing scatter plot shows a very weak positive relationship, meaning high Reading scores do not strongly predict high Writing scores in this dataset.

Dataset Overview and Data Cleaning

The dataset contains student-level academic performance records. Each row represents one student, while the columns describe demographic attributes, test preparation status, and subject performance.

Metric	Result
Number of students	100
Number of original columns	8
Duplicate rows	0
Duplicate Student IDs	0
Numeric score range	0 to 100
Test preparation categories after cleaning	Completed / Not Completed

Cleaning Actions Completed

- Checked all records for duplicate rows and duplicate Student IDs.
- Verified that Math, Reading, and Writing scores are numeric and within the expected 0-100 score range.
- Converted blank or missing test-preparation entries into a clear category: Not Completed.
- Created derived fields for Total Score, Average Score, Performance Level, Pass/Fail Status, Subject Strength, Subject Weakness, and Score Band.
- Prepared the data for Power BI reporting with clean categories and measurable KPIs.

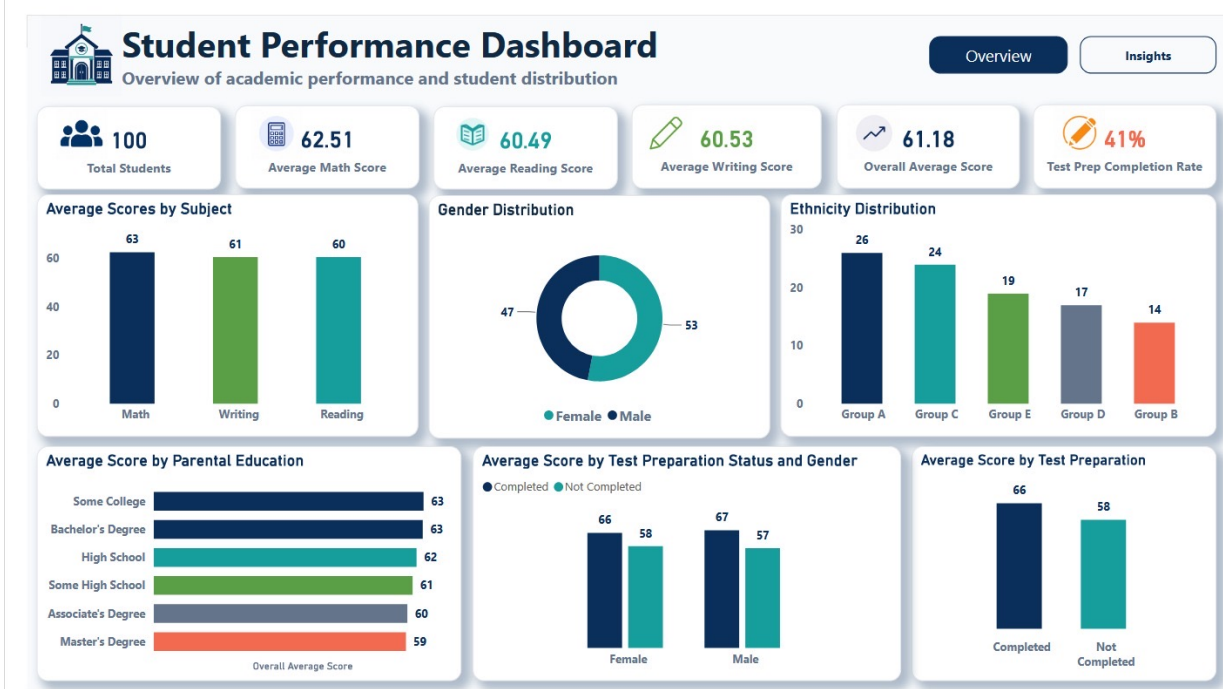
Key Calculated Columns and Measures

Name	Purpose
Total Score	Adds Math, Reading, and Writing scores for each student.
Average Score	Calculates the mean subject score for each student.
Performance Level	Groups students into achievement categories such as Good, Average, and Very Good.
Test Preparation Status	Replaces blank/None values with Not Completed for clearer reporting.
Subject Strength	Identifies the highest-scoring subject for each student.
Subject Weakness	Identifies the lowest-scoring subject for each student.
Score Band	Groups students into score bands such as 60-69, 50-59, and <50.
Test Prep Completion Rate	Measures the share of students who completed test preparation.

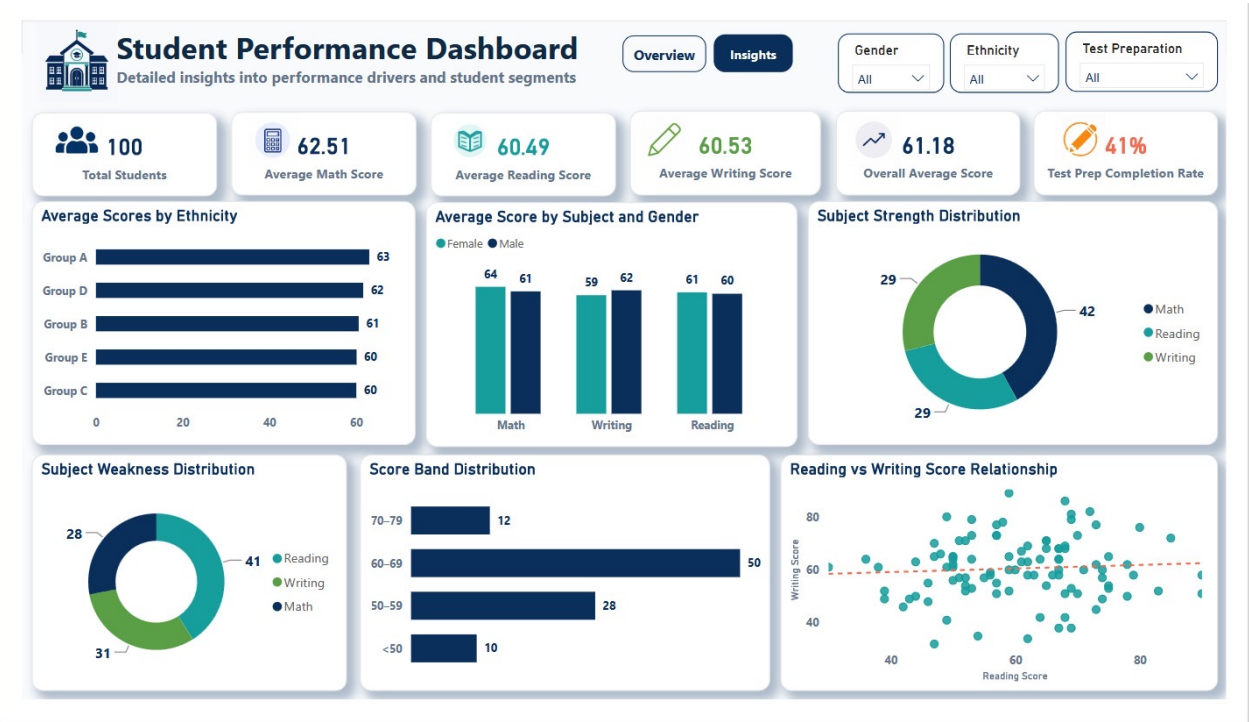
Power BI Dashboard Pages

The Power BI report was designed as a simple two-page dashboard. The first page provides a high-level overview of student performance and distribution. The second page focuses on performance drivers and student segment insights.

Dashboard Page 1 - Overview



Dashboard Page 2 - Insights



Dashboard 1 Analysis - Performance Overview

The Overview dashboard summarises the most important student performance indicators and provides a quick understanding of subject performance, gender composition, ethnicity distribution, parental education, and test-preparation impact.

KPI Summary

KPI	Value	Interpretation
Total Students	100	The dataset contains a manageable student sample for exploratory analysis.
Average Math Score	62.51	Math is the highest subject average, but only slightly higher than Reading and Writing.
Average Reading Score	60.49	Reading is marginally the lowest subject average.
Average Writing Score	60.53	Writing performance is close to Reading and Math, showing balanced subject performance.
Overall Average Score	61.18	Overall student performance is moderate.
Test Prep Completion Rate	41%	Less than half of students completed test preparation.

Chart Interpretations

Visual	Analysis
Average Scores by Subject	Math leads with approximately 63, followed by Writing at about 61 and Reading at about 60. The differences are small, meaning subject performance is balanced but not exceptional.
Gender Distribution	The dataset is fairly balanced by gender, with 53 female students and 47 male students. This helps avoid a heavily skewed gender sample.
Ethnicity Distribution	Group A has the highest number of students with 26, followed by Group C with 24. Group B has the fewest students with 14.
Average Score by Parental Education	Students from Some College and Bachelor's Degree backgrounds show the highest averages, while Master's Degree appears lowest in this sample. The variation is visible but not extremely large.
Average Score by Test Preparation Status and Gender	Both male and female students who completed test preparation scored higher than those who did not. This pattern supports the value of test preparation.
Average Score by Test Preparation	Completed test preparation students scored about 66 compared with 58 for Not Completed students, creating an 8.75-point advantage.

Dashboard 2 Analysis - Performance Insights

The Insights dashboard looks deeper into student segments and performance drivers. It focuses on ethnicity, gender-subject comparison, subject strengths and weaknesses, score bands, and the Reading/Writing relationship.

Segment and Distribution Analysis

Area	Finding
Average Score by Ethnicity	Group A has the highest average score at about 63. Group D follows at about 62, while Groups C and E are close to 60. The overall ethnicity performance gap is small.
Average Score by Subject and Gender	Female students are slightly higher in Math and Reading, while male students are slightly higher in Writing. The differences are not large enough to suggest a major gender-based performance gap.
Subject Strength Distribution	Math is the most common subject strength with 42 students, while Reading and Writing each appear as the strongest subject for 29 students.
Subject Weakness Distribution	Reading is the most common weakness, appearing as the lowest-scoring subject for 41 students. Writing follows with 31 students, and Math with 28.
Score Band Distribution	Most students are in the 60-69 band, with 50 students. Another 28 students are in the 50-59 band and 10 students are below 50.
Reading vs Writing Score Relationship	The correlation between Reading and Writing scores is approximately 0.07, indicating a very weak positive relationship in this sample.

Detailed Tables from the Analysis

Ethnicity	Avg Score	Parental Education	Avg Score
Group A	63.03	Some College	63.31
Group D	61.63	Bachelor's Degree	63.12
Group B	60.60	High School	61.67
Group E	60.11	Some High School	60.74
Group C	60.04	Associate's Degree	59.51
		Master's Degree	59.18

Additional Analytical Findings

The following tables provide supporting evidence behind the dashboard visuals and strengthen the report narrative.

Test Preparation Impact

Test Preparation Status	Students	Average Score
Completed	41	66.34
Not Completed	59	57.59

Students who completed test preparation outperformed students who did not by **8.75 points**. This is the largest and most actionable performance gap identified in the analysis.

Subject Strength and Weakness

Subject Strength	Students	Subject Weakness	Students
Math	42	Reading	41
Reading	29	Writing	31
Writing	29	Math	28

Score Band Distribution

Score Band	Students	Share of Total
70-79	12	12%
60-69	50	50%
50-59	28	28%
<50	10	10%

Recommendations

The recommendations below are based on the dashboard trends and computed measures. They are framed as practical actions a school, tutor, education administrator, or learning support team can apply.

Recommendation	Explanation
1. Expand test preparation participation	Only 41% of students completed test preparation, but completed students scored 8.75 points higher on average. Encourage wider participation through early reminders, guided sessions, and targeted support for students who have not completed preparation.
2. Target students below 60	A total of 38 students fall below 60: 28 in the 50-59 band and 10 below 50. These students should receive priority support through remedial lessons, study plans, mentoring, and progress tracking.
3. Strengthen Reading support	Reading is the most common subject weakness for 41 students. Even though the average Reading score is close to the other subjects, targeted reading comprehension and vocabulary support may help many students.
4. Support the large mid-performing group	Fifty students are in the 60-69 band. Helping this group move into the 70-79 band could significantly improve overall academic performance.
5. Use parental education insights carefully	The performance gap across parental education groups is moderate. Use this as a support signal, not a basis for assumptions. Provide learning resources and study guidance to all groups.
6. Collect more explanatory data	Future analysis would be stronger with attendance, study hours, assignment completion, teacher feedback, class level, prior scores, and socioeconomic indicators.

Limitations

- The dataset contains only 100 student records, so results should be interpreted as sample-based insights rather than universal conclusions.
- The dataset is cross-sectional. It does not include dates or historical performance, so trends over time cannot be evaluated.
- The analysis does not include attendance, study time, teacher quality, school resources, or socioeconomic factors, which may strongly influence academic outcomes.
- The correlation between Reading and Writing is weak in this sample, but a larger dataset may reveal a different pattern.
- Dashboard values are useful for descriptive analysis, but they do not prove causation.

Conclusion

The Student Performance Dashboard provides a clear and client-ready view of academic outcomes across 100 students. The strongest actionable insight is that test preparation is associated with higher scores across both male and female students. The dashboard also identifies students below 60 as a key support group, shows Reading as the most common subject weakness, and highlights the need to move the large 60-69 score-band population into higher performance bands.

Overall, the project demonstrates how student data can be converted into meaningful education insights using data cleaning, calculated fields, Power BI visuals, and a structured analytical narrative. The dashboard is suitable for monitoring academic performance, identifying support needs, and guiding data-driven learning interventions.

Suggested Next Steps

- Add a student support tracking page showing students below 60 and follow-up actions.
- Add a time dimension if future assessment records become available.
- Compare performance before and after test-preparation sessions.
- Create automated alerts for students moving into the <50 or 50-59 bands.

Appendix - Key Power BI Logic

The following sample DAX logic was used to prepare the report fields and measures in Power BI.

Total Score =

```
'Student Performance'[Math Score] +  
'Student Performance'[Reading Score] +  
'Student Performance'[Writing Score]
```

Average Score = DIVIDE('Student Performance'[Total Score], 3, 0)

Test Preparation Status =

```
IF('Student Performance'[Test Preparation] = "None" ||  
    ISBLANK('Student Performance'[Test Preparation]),  
    "Not Completed",  
    'Student Performance'[Test Preparation])
```

Overall Average Score = AVERAGE('Student Performance'[Average Score])

Test Prep Completion Rate =

```
DIVIDE(  
    CALCULATE(COUNTROWS('Student Performance'),  
        'Student Performance'[Test Preparation Status] = "Completed"),  
    COUNTROWS('Student Performance'),  
    0  
)
```

Score Band =

```
SWITCH(TRUE(),  
    'Student Performance'[Average Score] >= 80, "80-100",  
    'Student Performance'[Average Score] >= 70, "70-79",  
    'Student Performance'[Average Score] >= 60, "60-69",  
    'Student Performance'[Average Score] >= 50, "50-59",  
    "<50"  
)
```